

Adaptive Retention Under Continual Learning

Rate, reinforcement-strength, and task-sequence ablations for selective memory dynamics

Tier 2 - Master's Research Design, revised as the next experimental extension

Adaptive Memory Dynamics Research Program Stage One revision - September 2026

Evidence status: Stage One is a reproducible sklearn-digits-8x8 experiment across five independent seeds. Reinforced retention approximately matches the persistent control on ordinary accuracy and Task B adaptation; passive decay reduces accuracy; neither decay mechanism improves Task A retention in the current sequential setup.

Abstract

The graduate extension treats forgetting rate and reinforcement strength as experimental variables rather than fixed design choices. Building from the mixed Stage One result, the study maps the retention-adaptation tradeoff across mechanism hyperparameters, task sequences, and benchmark families. The objective is not to force adaptive forgetting to outperform persistent retention, but to determine whether reproducible operating regimes exist in which selective decay improves adaptation, generalization, or memory efficiency without excessive loss of prior-task competence.

1. Stage One motivation

The current evidence creates a concrete graduate question: reinforced retention approximately matches the persistent baseline on ordinary accuracy and Task B performance, yet the persistent baseline retains Task A better. Passive decay performs worse on both ordinary accuracy and prior-task retention. A graduate study should therefore identify whether these outcomes are structural or parameter-specific.

2. Experimental factors

Factor	Planned treatment
Decay rate	multi-level sweep including zero
Reinforcement strength	multi-level sweep including no reinforcement
Activation threshold	controlled ablation
Idle duration	short-to-long inactivity intervals
Task sequence	multiple class/task orderings
Benchmark	digits validation then larger planned datasets
Seeds	independent replicated training runs

Figure 1. Hypothesis schematic only. The response shape is illustrative; the optimum, if any, must be estimated empirically.

3. Outcomes

- Prior-task retention and forgetting.
- New-task adaptation accuracy and learning speed.
- Ordinary held-out generalization.
- Post-idle stability.
- Weight/representation drift and retention statistics.
- A pre-defined retention-adaptation utility or Pareto frontier rather than a single cherry-picked metric.

4. Analysis plan

The principal analysis estimates mechanism-by-rate and mechanism-by-task interactions with seed-level uncertainty. Hyperparameter selection and final evaluation should be separated so that the same seeds are not used both to discover and confirm a favorable setting. Results should be reported as surfaces or Pareto frontiers when retention and adaptation trade off.

5. Contribution boundary

This tier studies engineered memory dynamics inspired by cognitive principles; it does not claim to reproduce human memory. Similarity to exponential forgetting or reinforcement-sensitive retention is a modeling analogy unless biological data are explicitly fitted and validated.

Figures

